

Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
9

Appdev1

Section Id :	64065375365
Section Number :	9
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	30
Number of Questions to be attempted :	30
Section Marks :	100
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653159803
Question Shuffling Allowed :	No

Question Number : 177 Question Id : 6406531039215 Question Type : MCQ
Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MODERN APPLICATION DEVELOPMENT I (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE [TOP](#) FOR THE SUBJECTS REGISTERED BY YOU)

Options :
6406533511044. ✓ YES

6406533511045. ✖ NO

Sub-Section Number :	2
Sub-Section Id :	640653159804
Question Shuffling Allowed :	Yes

Question Number : 178 Question Id : 6406531039223 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Consider the following models Library and Book corresponding to tables library and book in SQLite database.

```
class Library(db.Model):
    id = db.Column(db.Integer(), primary_key = True)
    name = db.Column(db.String(), unique = True)

class Book(db.Model):
    id = db.Column(db.Integer(), primary_key = True)
    name = db.Column(db.String(), unique = True)
    library = db.Column(db.Integer(), unique = True, db.ForeignKey("section.id"))
```

Based on the model schemas, what relationship do the classes Library and Book share?

Options :

6406533511074. ✖ Many-to-Many

6406533511075. ✖ One-to-Many

6406533511076. ✖ One-to-One

6406533511077. ✔ The tables are not at all related

Question Number : 179 Question Id : 6406531039224 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Consider the following python code snippet “app.py”, the HTML files, “base.html” and “home.html” residing in “templates” folder.

app.py

```
from flask import Flask, render_template
app = Flask(__name__)
@app.route('/')
def home():
    return render_template('home.html')
app.run(debug=True)
```

home.html

```
{% extends "base.html" %}
{% block content %}
<h3>This is from home.html</h3>
{% endblock %}
```

base.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <title>Document</title>
</head>
<body>
    <h3 style="color: blue;"> This is from base.html </h3>
    {% block content %}
    {% endblock %}
</body>
</html>
```

What will be the rendered output for base URL if flask app is running locally on <http://localhost:5000>

Options :

6406533511078. ✖ **This is from home.html**

This is from base.html

6406533511079. ✔ **This is from home.html**

This is from home.html

6406533511080. ✖ **This is from base.html**

6406533511081. ✖ Error

Question Number : 180 Question Id : 6406531039225 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Consider the following Python code snippet.

```
from string import Template

sentence = Template("A $var1 fox $var2 over the $var3 dog")

output = sentence.safe_substitute(word_dict)

print(output)
```

Which of the following options correctly represent(s) the dictionary `word_dict`, such that the code does not throw any error when run in the terminal?

Options :

6406533511082. ✖ `word_dict = {'var1': 'quick'}`

6406533511083. ✖ `word_dict = {'var1': 'quick', 'var2': 'jumps'}`

6406533511084. ✖ `word_dict = {'var1': 'quick', 'var2': 'jumps', 'var3': 'lazy'}`

6406533511085. ✔ All of these

Question Number : 181 Question Id : 6406531039245 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Which of the following is a time series database?

Options :

6406533511145. ✖ MongoDB

6406533511146. ✔ InfluxDB

6406533511147. ✖ MySQL

6406533511148. ✖ PostgreSQL

Question Number : 182 Question Id : 6406531039247 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Match the platforms given in column A with their correct features in column B.

Column A	Column B
1. Desktop	A. Data center storage
2. Mobile	B. Keyboard
3. Web-based	C. Single function
4. Embedded	D. tilt

Options :

6406533511153. ✖ 1 - A, 2 - B, 3 - C, 4 - D

6406533511154. ✖ 1 - D, 2 - A, 3 - B, 4 - C

6406533511155. ✔ 1 - B, 2 - D, 3 - A, 4 - C

6406533511156. ✖ 1 - A, 2 - C, 3 - D, 4 - B

Sub-Section Number :

3

Sub-Section Id :

640653159805

Question Shuffling Allowed :

Yes

Question Number : 183 Question Id : 6406531039218 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Consider the following view function.

```
from flask import Flask, request
app = Flask(__name__)

@app.route('/employee', methods=['GET', 'POST'])
def show_info():
    info = request.args
    details = {
        'Department': info['dept'],
        'ID': info['id'],
        'Role': info['role']
    }
    return details

app.run()
```

If this flask app is running locally on `http://127.0.0.1:5000`, which of the following URLs will be handled by the controller correctly?

Options :

6406533511054. ✖ `http://127.0.0.1:5000?dept=HR&id=101&role=manager`

6406533511055. ✖ `http://127.0.0.1:5000/HR/101/manager`

6406533511056. ✔ `http://127.0.0.1:5000/employee?dept=HR&id=101&role=manager`

6406533511057. ✖ `http://127.0.0.1:5000/employee/HR/101/manager`

Question Number : 184 Question Id : 6406531039221 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

A certain internet connection is established such that its bandwidth reduces to half of its current value every hour. A user connects to this connection at 01:00 pm with the network speed of 4 Mbps, how much data (in GBs) will be consumed by the user by 04:00 p.m in the evening? [round off upto 2 decimal places](Assume that the user is using the entire bandwidth)

Options :

6406533511066. ✖ 1.35

6406533511067. ✖ 2.7

6406533511068. ✔ 3.15

6406533511069. ✖ 5.4

Question Number : 185 Question Id : 6406531039222 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Consider the following function to be tested and test functions given in the Python code snippet below.

file.py

```
def sample_function(x):  
    var = []  
    for i in range(x):  
        var.append((i)**3)  
    return var  
  
def test_func1():  
    assert 125 in sample_function(6)  
  
def test_func2():  
    assert 64 in sample_function(8)  
  
def sample_test3():  
    assert 125 in sample_function(5)
```

What will be the summary of the output for the command
pytest file.py on the terminal?

Options :

6406533511070. ✖ ===== 1 failed, 2 passed in 0.06s =====

6406533511071. ✖ ===== 2 failed, 1 passed in 0.06s =====

6406533511072. ✖ ===== 3 passed in 0.06s =====

6406533511073. ✔ ===== 2 passed in 0.06s =====

Question Number : 186 Question Id : 6406531039230 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Consider the following HTML document.

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Document</title>
  <script>
    function makeBold(){
      //MISSING JS CODE
    }
  </script>
</head>
<body>
  Enter your name : <input type="text" id="txt1" name="txt1">
  <br>
  <input type="button" onclick="makeBold();" value="Click Me">
</body>
</html>

```

Replace the **MISSING JS CODE** part with the correct JS code that will change the text box content to bold?

Options :

6406533511099. ✖

```

t=document.getElementById("txt1").value;
t.style.fontWeight="bold";

```

6406533511100. ✖

```

t=document.getElementById("txt1").value;
t.style.fontWeight="bold";

```

6406533511101. ✖

```

t=document.getElement("txt1");
t.style.fontWeight="bold";

```

6406533511102. ✔

```

t=document.getElementById("txt1");
t.style.fontWeight="bold";

```

Question Number : 187 Question Id : 6406531039242 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Match the following terms with their correct descriptions:

A. DOM Event	1. The structure that represents the HTML document and allows interaction with it.
B. Attributes	2. A method or action triggered by user interactions like clicks, typing, etc.
C. Document	3. The individual elements or parts of a page, such as elements, attributes, or text.
D. Node	4. The properties that define the characteristics of an HTML element, like <code>id</code> , <code>class</code> , etc.
E. DOM (Document Object Model)	5. A programming interface for web documents that provides structured access to the document's elements.

Options :

6406533511133. ✓ A → 2, B → 4, C → 1, D → 3, E → 5

6406533511134. ✗ A → 3, B → 4, C → 1, D → 2, E → 5

6406533511135. ✗ A → 2, B → 4, C → 5, D → 3, E → 1

6406533511136. ✗ A → 3, B → 4, C → 5, D → 2, E → 1

Question Number : 188 Question Id : 6406531039244 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Which of the following is an example of an application using Attribute-Based Access Control (ABAC)?

Options :

6406533511141. ✗ A system where users are granted access based on their job roles within an organization.

6406533511142. ✓ A cloud service that grants or restricts access to files based on attributes such as user department, job title, or current location.

6406533511143. ✗ A file-sharing system where the file owner decides who can access their documents.

6406533511144. ✗ A system that enforces access based on security classifications and clearance levels.

Question Number : 189 Question Id : 6406531039249 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Consider the following Table products in SQLite database created using model class Product and the python file main.py given below.

products:

id	pro_name	section
1	Palm Oil	food
2	Butter	food
3	Pencil	art
4	Face wash	cosmetics
5	Olive	vegetables

main.py

```
from jinja2 import Template
templates = """
    {% if result %}
    <ol>
        {% for product in products %}
            <li>{{product.pro_name}}</li>
            <li>{{product.section}}</li>
        {% endfor %}
    </ol>
    {% endif %}
    """

products = Product.query.filter_by(section = 'food').all()

Templates = Template(templates)
result = Templates.render(products = products)
print(result)
```

What will be the output on the terminal?

Options :

```
Palm Oil food
Butter food
```

6406533511161. ✖

```
<ol>
    <li>Palm Oil food</li>
    <li>Butter food</li>
</ol>
```

6406533511162. ✖

6406533511163. ✖

Palm Oil
food
Butter
food

6406533511164. ✔

```
<ol>  
  <li>Palm Oil</li>  
  <li>food</li>  
  <li>Butter</li>  
  <li>food</li>  
</ol>
```

Question Number : 190 Question Id : 6406531039250 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Consider the following Python code snippet.

```
from math import factorial  
  
def prettify(func):  
    def wrapper(num):  
        try:  
            return func(num)  
        except:  
            return "Please check your number!"  
    return wrapper  
  
@prettify  
def fact(num):  
    return factorial(num)  
  
out = fact(5)  
print(out)
```

What will the output if the above Python code is run on the terminal?

Options :

6406533511165. ✖ -120

6406533511166. ✔ 120

6406533511167. ✖ Please check your number!

6406533511168. ✖ None of these.

Sub-Section Number : 4
Sub-Section Id : 640653159806
Question Shuffling Allowed : Yes

Question Number : 191 Question Id : 6406531039216 Question Type : MCQ

Correct Marks : 4.5

Question Label : Multiple Choice Question

Consider the following Python code snippet.

file.py

```
from string import Template as makeTemplate
from jinja2 import Template
import sys

var = sys.argv[2]

data = {"name": "Alice", "activity": "data analysis",
        "tool": "Python", "goal": "insights"}

temp = "{{name}} performs $activity using $tool to gain {{goal}}."

if var == "python":
    temp = makeTemplate(temp)
    out = temp.substitute(data)
    output = Template(out)
    output = output.render(data)
    print(output)
else:
    temp = Template(temp)
    output = temp.render(data)
    print(output)
```

What will be printed on the terminal for the command `python file.py ml python` ?

Options :

6406533511046. ✖ Alice performs \$activity using \$tool to gain insights.

6406533511047. ✖ {{name}} performs \$activity using \$tool to gain {{goal}}.

6406533511048. ✔ Alice performs data analysis using Python to gain insights.

6406533511049. ✖

Alice performs \$activity using \$tool to gain {{goal}}.

Question Number : 192 Question Id : 6406531039219 Question Type : MCQ

Correct Marks : 4.5

Question Label : Multiple Choice Question

Consider the following HTML document.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Document</title>
  <style>
    p {
      background-color: yellow;
      color: red;
      display: inline-block;
    }
    #id {
      border: 2px solid purple;
      color:blue ;
    }
    .class1 {
      background-color: aqua;
      color: red;
    }
    .class2{
      background-color: lightgreen;
      color: darkgreen;
    }
  </style>
</head>
<body>
  <p class="class1" id="id">Code 1</p>
  <p class="class1 class2" >Code 2</p>
  <p class="class1">Code 3</p>
  <p >Code 4</p>
</body>
</html>
```

How will the browser render above HTML file?

Options :

6406533511058. ✖

Code 1
Code 2
Code 3
Code 4

Code 1
Code 2
Code 3
Code 4

6406533511059. ✖

6406533511060. ✖

Code 1 Code 2 Code 3 Code 4

6406533511061. ✔

Code 1 Code 2 Code 3 Code 4

Question Number : 193 Question Id : 6406531039246 Question Type : MCQ

Correct Marks : 4.5

Question Label : Multiple Choice Question

Refer to the project with three files: `conftest1.py`, `test_multiples_of_3_and_6.py`, and `test_multiples_of_13.py`.

The purpose of the tests is to verify if numbers generated by each function are divisible by 39. However, there is a setup issue in this project.

- `conftest1.py` contains a fixture that provides test data.
- `test_multiples_of_3_and_6.py` has two functions: one checks divisibility by 3, and the other by 6.
- `test_multiples_of_13.py` contains a function to check divisibility by 13.

Here is the content of each file:

`conftest1.py`

```
import pytest

@pytest.fixture
def divisible_by_39():
    return 39 # Fixture providing the number to be tested
```

`test_multiples_of_3_and_6.py`

```
import pytest

def test_divisible_by_3(divisible_by_39):
    assert divisible_by_39 % 3 == 0 # Checks if divisible by 3

def test_divisible_by_6(divisible_by_39):
    assert divisible_by_39 % 6 == 0 # Checks if divisible by 6
```

`test_multiples_of_13.py`

```
import pytest

def test_divisible_by_13(divisible_by_39):
    assert divisible_by_39 % 13 == 0 # Checks if divisible by 13
```

When you run `pytest -k divisible`, which of the following is **true**?

Options :

6406533511149. ✖ ===== 1 passed in 0.02s =====

6406533511150. ✖ ===== 3 passed in 0.02s =====

6406533511151. ✖ ===== 1 failed, 2 passed in 0.06s =====

6406533511152. ✔ ===== 3 errors in 0.02s =====

Question Number : 194 Question Id : 6406531039248 Question Type : MCQ

Correct Marks : 4.5

Question Label : Multiple Choice Question

Consider the following Python code snippet that returns the potential energy stored in an object of a certain mass at a certain height. (PE = mgh, where "g" is acceleration due to gravity).

```
from jinja2 import Template

x = Template(template)
```

What should be the value of template if,

```
print(x.render(mass = "5", height = "22")) returns 1100 and
print(x.render()) returns 3000
```

Options :

6406533511157. ✓ `"{{ mass|default(3)|int*gravity|default(10)|int*height|default(100)|int }}"`
6406533511158. ✗ `"{{ mass|int|default(3)*gravity|int|default(10)*height|int|default(100) }}"`
6406533511159. ✗ `"{{ mass|default(5)|int*gravity|default(10)|int*height|default(100)|int }}"`
6406533511160. ✗ `"{{ mass|int|default(5)*gravity|int|default(10)*height|int|default(100) }}"`

Sub-Section Number :

5

Sub-Section Id :

640653159807

Question Shuffling Allowed :

Yes

Question Number : 195 Question Id : 6406531039217 Question Type : MSQ

Correct Marks : 4.5 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider the following flask app. Given that `test_request_context()` allows text to be printed on the terminal, which of the following statements is/are correct?

```
from flask import Flask, url_for
app = Flask(__name__)

@app.route('/products')
def products():
    return 'Product List'

@app.route('/product/<int:product_id>/<string:name>')
def product_detail(product_id, name):
    return f'Product ID: {product_id}, Name: {name}'

with app.test_request_context():
    #== print statement ==#
```

Options :

If `#== print statement ==#` is replaced by:
`print(url_for('products', category="electronics"))`,
the output on the terminal will be;

6406533511050. ✓ `/products?category=electronics`

If `#== print statement ==#` is replaced by:
`print(url_for('products', category="electronics"))`,
the output on the terminal will be;

6406533511051. ✗ `/products/electronics`

If `#== print statement ==#` is replaced by:
`print(url_for('product_detail', product_id=101, name="Gadget", discount=20))`,
the output on the terminal will be;

6406533511052. ✓ `/product/101/Gadget?discount=20`

If `#== print statement ==#` is replaced by:
`print(url_for('product_detail', product_id=101, name="Gadget", discount=20))`,
the output on the terminal will be;

6406533511053. ✗ `/product/101/Gadget/20`

Sub-Section Number :

6

Sub-Section Id :

640653159808

Question Shuffling Allowed :

Yes

Question Number : 196 Question Id : 6406531039220 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider the following code snippet.

```
from flask import Flask
import logging
app = Flask(__name__)

logging.basicConfig(filename='logger.log', level=logging.INFO,
format=f'%(asctime)s : %(message)s')

@app.route('/home')
def home():
    app.logger.debug('This is a debug log')
    app.logger.warning('This is a warning log')
    return f"Application Dashboard"

if __name__ == "__main__":
    app.run()
```

Suppose the given flask application is running locally on `http://127.0.0.1:5000`, which of the following statements is/are true if you hit the URL `http://127.0.0.1:5000/home` ?

Options :

- 6406533511062. ✓ A file called "logger.log" will be created inside the Flask application root folder.
- 6406533511063. ✗ The message "Application Dashboard" will be displayed within the log file.
- 6406533511064. ✗ The message "This is a debug log" will be logged inside the "logger.log" file.
- 6406533511065. ✓ The message "This is a warning log" will be logged inside the "logger.log" file.

Question Number : 197 Question Id : 6406531039226 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider the following flask application running on the base URL and is accessed through a browser. Select the correct option(s).


```

app = Flask(__name__)

users = {
    1 : {"Name": "Mukesh", "role": "Admin", "access": True},
    2 : {"Name": "Gautam", "role": "User", "access": True},
    3 : {"Name": "Narendra", "role": "Admin", "access": True},
    4 : {"Name": "Amit", "role": "User", "access": False}
}

@app.route('/login')
def auth():
    cred = request.args
    if users[int(cred["id"])].get("access"):
        id = int(cred["id"])
        user = users[id]
        return "Welcome, " + user.get("Name") + ", you have " + user.get("role")
+ " access"
    else:
        abort(403)

@app.errorhandler(403)
def no_access(error):
    return "You are an unauthorized user!"

app.run(debug = True)

```

Options :

For the URL: `http://127.0.0.1:5000?id=2`, The browser will render;

6406533511086. ✖ Welcome, Gautam, you have User access.

For the URL: `http://127.0.0.1:5000/login?id=1`, The browser will render;

6406533511087. ✔ Welcome, Mukesh, you have Admin access.

For the URL: `http://127.0.0.1:5000/login?id=3`, The browser will render;

6406533511088. ✖ Welcome, Narendra, you have User access.

For the URL: `http://127.0.0.1:5000/login?id=4`, The browser will render;

6406533511089. ✔ You are an unauthorized user!.

Question Number : 198 Question Id : 6406531039238 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider the following Python code snippet.

File name: **main.py**

```
import sys

def func1(x):
    return x > 0

def func2(x):
    return x % 2 == 0

def test_fun3(x):
    assert func1(x) == True, str(x) + " is not even/odd"
    assert func2(x) == True, str(x) + " is even number"
    assert x % 2 != 0, str(x) + " is odd number"

x = int(sys.argv[1])
test_fun3(x)
```

Which of the following is the correct combinations of “**command line argument**” and “**assertion error**”?

Options :

6406533511120. ✓ *Command: python main.py 9
AssertionError: 9 is even number*

6406533511121. ✓ *Command: python main.py 90
AssertionError: 90 is odd number*

6406533511122. ✓ *Command: python main.py 0
AssertionError: 0 is not even/odd*

6406533511123. ✗ *Command: python main.py 9
AssertionError: 9 is odd number*

6406533511124. ✗ *Command: python main.py 90
AssertionError: 90 is even number*

Question Number : 199 Question Id : 6406531039243 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following statements about the Constraint Validation API is/are true?

Options :

6406533511137. ✔ The Constraint Validation API ensures that form input meets client-side validation rules before submission.

6406533511138. ✖ The Constraint Validation API is sufficient to prevent server-side attacks like SQL injection.

6406533511139. ✔ It is possible to bypass the Constraint Validation API by directly sending requests to the server.

6406533511140. ✖ The Constraint Validation API automatically validates all data on the server side.

Sub-Section Number :

7

Sub-Section Id :

640653159809

Question Shuffling Allowed :

Yes

Question Number : 200 Question Id : 6406531039228 Question Type : MSQ

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following statements is invalid in the context of the primary key?

Options :

6406533511091. ✔ The primary key can be null

6406533511092. ✖ The primary key can be auto-incremented

6406533511093. ✔ The primary key allows duplicate values

6406533511094. ✖ The primary key can be referenced by foreign keys in other tables.

Question Number : 201 Question Id : 6406531039229 Question Type : MSQ

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider the following HTML document.

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>IITM</title>
</head>
<body>
  <h2>Welcome to IITM </h2>
  <h3>Online Degree</h3>
  Courses offered
  <ol>
    <li>Foundation</li>
    <ul>
      <li>CT</li>
      <li>Python</li>
      <li>Maths-I</li>
    </ul>
    <li>Diploma</li>
    <ul>
      <li>PDSA</li>
      <li>MAD-I</li>
      <li>MAD-II</li>
    </ul>
  </ol>
</body>
</html>

```

Which of the following statements is/are correct about the above code snippet?

Options :

6406533511095. ✔ List Items, Foundations and Diploma are numbered 1 and 2
6406533511096. ✖ List Items, Foundations and Diploma are bulleted
6406533511097. ✖ List Items, PDSA, MAD-I and MAD-II are numbered 1, 2 and 3
6406533511098. ✔ List Items, PDSA, MAD-I and MAD-II are bulleted

Sub-Section Number :

8

Sub-Section Id :

640653159810

Question Shuffling Allowed :

Yes

Question Number : 202 Question Id : 6406531039227 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Consider a comma-separated values (CSV) file with all characters set ["0", "1", "2", "3", "4", "5", "6", "7", "8", "9", "a", "b"]. You need to store a file consisting of 2000 characters mentioned only in the above list without spaces. How many minimum number of bytes are required to store this CSV file efficiently?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

1000

Question Number : 203 **Question Id :** 6406531039234 **Question Type :** SA

Correct Marks : 3

Question Label : Short Answer Question

You are connected to a server that can process 1500 requests per second. Each request is a constant size of 100KB. You are connected to the above server through a network line with a speed of 2Gbps. What is the maximum number of requests per second that the server can handle(in KBps)? (Consider that 1000Bytes=1KB, 1000KB = 1MB)?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

150000

Sub-Section Number : 9

Sub-Section Id : 640653159811

Question Shuffling Allowed : No

Question Id : 6406531039231 **Question Type :** COMPREHENSION **Sub Question Shuffling**

Allowed : No **Group Comprehension Questions :** No **Question Pattern Type :** NonMatrix

Question Numbers : (204 to 205)

Question Label : Comprehension

Consider the following Python flask code running on the url `http://127.0.0.1:5000`.

```
from flask import Flask

app = Flask(__name__)
movies = {
    "Spider man1": ["action", 5, "Sam Raimi"],
    "Batman": ["action", 3, "Matt Reeves"],
    "Joker": ["thriller", 4, "Todd Philips"],
    "Ted": ["comedy", 3, "Seth MacFarlane"],
}

def doSomething(g):
    l = []
    for key in movies:
        if movies[key][0] == g:
            l.append(key)
    return l

@app.route("/<genre>")
def home(genre):
    return doSomething(genre)

if __name__ == "__main__":
    app.run(debug=True)
```

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 204 Question Id : 6406531039232 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Which of the following urls will display `Joker` on the browser?

Options :

6406533511103. ✖ `http://127.0.0.1:5000/action`

6406533511104. ✖ `http://127.0.0.1:5000/thriller`

6406533511105. ✔ `http://127.0.0.1:5000/comedy`

6406533511106. ✖ `http://127.0.0.1:5000?genre=thriller`

Question Number : 205 Question Id : 6406531039233 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

What will be displayed on the browser for the url:
`http://127.0.0.1:5000?genre=action` ?

Options :

6406533511107. ✖ [5, 3]

6406533511108. ✔ ["Spider man1", "Batman"]

6406533511109. ✖ ["Sam Raimi", "Matt Reeves"]

6406533511110. ✖ None of these.

Sub-Section Number :	10
Sub-Section Id :	640653159812
Question Shuffling Allowed :	No

Question Id : 6406531039235 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (206 to 207)
Question Label : Comprehension

Consider the below Python flask code and answer the given subquestions if the app is running on "<http://127.0.0.1:5000/>"

File: **app.py**

```
app = Flask(__name__)
api = Api(app)

accounts_data = {
    "sb101": {"name": "Raj", "balance": 5600},
    "ca101": {"name": "David", "balance": 7600},
    "fd101": {"name": "Jack", "balance": 71600},
    "sb103": {"name": "Syam", "balance": 9600},
}

p = reqparse.RequestParser()
p.add_argument("balance")

class Error(HTTPException):
    def __init__(self, status, error):
        self.response = make_response({"Error": error}, status)

class Accounts(Resource):
    def get(self, id):
        if id in accounts_data:
            acct = accounts_data[id]
            return acct
        else:
            error = "Sorry, account no. " + id + " does not exist!"
            raise Error(404, error)

    def put(self, id):
        args = p.parse_args()
        if id in accounts_data:
            accounts_data[id]["balance"] = args["balance"]
            return accounts_data[id]
        else:
            error = "Account no. " + id + " does not exist and balance not updated..."
            raise Error(404, error)

api.add_resource(Accounts, "/api/get_account/<id>", "/api/update_balance/<id>")
app.run(debug=True)
```

Sub questions

Question Number : 206 Question Id : 6406531039236 Question Type : MCQ

Correct Marks : 4.5

Question Label : Multiple Choice Question

What will be the output of the below terminal request?

```
curl http://127.0.0.1:5000/api/get_account/fd101 -X GET
```

Options :


```
{  
    "name": "Jack",  
    "balance": 71600  
}
```

6406533511112. ✓

6406533511113. ✗ Sorry, account no. fd101 does not exist

6406533511114. ✗ Account no. fd101 does not exist and balance not updated...

```
{  
    "name": "Syam",  
    "balance": 9600  
}
```

6406533511115. ✗

Question Number : 207 Question Id : 6406531039237 Question Type : MCQ

Correct Marks : 4.5

Question Label : Multiple Choice Question

Which of the following command will
update balance of account no. **ca101**?

Options :

6406533511116. ✗ `curl http://127.0.0.1:5000/api/update_balance/ca101 -X POST -d {"balance":"15890"} -H "Content-Type: application/json"`

6406533511117. ✗ `curl http://127.0.0.1:5000/api/get_balance/ca101 -X POST -d {"balance":"15890"} -H "Content-Type: application/json"`

6406533511118. ✗ `curl http://127.0.0.1:5000/api/get_balance/ca101 -X PUT -d {"balance":"15890"} -H "Content-Type: application/json"`

6406533511119. ✓ `curl http://127.0.0.1:5000/api/update_balance/ca101 -X PUT -d {"balance":"15890"} -H "Content-Type: application/json"`

Sub-Section Number :

11

Sub-Section Id :

640653159813

Question Shuffling Allowed :

No

Question Id : 6406531039239 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (208 to 209)

Question Label : Comprehension

Use the Python code given below for answering the subquestions:

```
from flask import Flask, jsonify, request, make_response
from flask_restful import Api, Resource

app = Flask(__name__)
api = Api(app)

items = []

class Item(Resource):
    def get(self, name):
        item = None
        for i in items:
            if i['name'] == name:
                item = i
                break
        return make_response(jsonify(item) if item else ('', 404))

    def post(self, name):
        # Check if item already exists
        item_exists = False
        for i in items:
            if i['name'] == name:
                item_exists = True
                break
        if item_exists:
            return make_response(jsonify({'message': 'Item already exists'}),
400)

        data = request.get_json()
        new_item = {'name': name, 'price': data['price']}
        items.append(new_item)
        return make_response(jsonify(new_item), 201)

api.add_resource(Item, '/item/<string:name>')

if __name__ == '__main__':
    app.run(debug=True)
```

Sub questions

Question Number : 208 Question Id : 6406531039240 Question Type : MCQ

Correct Marks : 4.5

Question Label : Multiple Choice Question

Assume you have the following Item API endpoint as shown in the code. If a client sends a POST request with an existing item name but missing the price field in the JSON payload, what will happen, and why?

Options :

6406533511125. ✖ The server will add a new item with **None** as the price value and return a 201 status code since there is no validation for the price field.

6406533511126. ✔ The server will raise a **KeyError** because it tries to access `data['price']` without validating if price is present, returning a 500 Internal Server Error.

6406533511127. ✖ The server will return a 400 status code with the message **"Item already exists"** because it first checks for the item's existence before attempting to access price.

6406533511128. ✖ The server will reject the request and return a 404 status code, indicating a missing field, because price is required.

Question Number : 209 Question Id : 6406531039241 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

What will be the status code of the response for the request:

curl <http://127.0.0.1:5000/item/candle?price=120> -X POST

Options :

6406533511129. ✖ 201

6406533511130. ✖ 404

6406533511131. ✖ 500